Лабораторная работа №4

Операционные системы

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Информация

Докладчик

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Цель работы

Получение навыков продвинутой работы с репозиториями git и релизами.

Задание

1. Выполнить работу для тестового репозитория.
2. Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

Теоретическое введение

Gitflow Workflow опубликована и популяризована Винсентом Дриссеном. Gitflow Workflow предполагает выстраивание строгой модели ветвления с учётом выпуска проекта. Данная модель отлично подходит для организации рабочего процесса на основе релизов.v Работа по модели Gitflow включает создание отдельной ветки для исправлений ошибок в рабочей среде.

Семантическое версионирование описывается в манифесте семантического версионирования. Кратко его можно описать следующим образом: Версия задаётся в виде кортежа МАЖОРНАЯ_ВЕРСИЯ.МИНОРНАЯ_ВЕРСИЯ.ПАТЧ. Номер версии следует увеличивать: МАЖОРНУЮ версию, когда сделаны обратно несовместимые изменения API. МИНОРНУЮ версию, когда вы добавляете новую функциональность, не нарушая обратной совместимости. ПАТЧ версию, когда вы делаете обратно

Выполнение лабораторной работы

Устанавливаю nodejs, пакетный менеджер для него pnpm и gitflow.

```
gracimildevnanque@fedora:~$ sudo dnf install nodejs pnpm
[sudo] senha para gracimildevnanque:
Updating and loading repositories:
Repositories loaded.
Package                          Arch       Version                          Repository          Size
Installing:
 nodejs                          x86_64     1:22.22.0-2.fc42                 updates          161.6 KiB
 pnpm                            noarch     10.9.0-1.fc42                    updates           16.3 MiB
Installing dependencies:
 nodejs-libs                     x86_64     1:22.22.0-2.fc42                 updates           78.1 MiB
Installing weak dependencies:
 nodejs-docs                     noarch     1:22.22.0-2.fc42                 updates           97.5 MiB
 nodejs-full-i18n                x86_64     1:22.22.0-2.fc42                 updates           30.4 MiB
 nodejs-npm                      x86_64     1:10.9.4-1.22.22.0.2.fc42        updates            9.2 MiB

Transaction Summary:
 Installing:        6 packages

Total size of inbound packages is 43 MiB. Need to download 43 MiB.
After this operation, 232 MiB extra will be used (install 232 MiB, remove 0 B).
Is this ok [y/N]: y
[1/6] nodejs-1:22.22.0-2.fc42.x86_64                          100% | 192.6 KiB/s |  51.4 KiB |  00m00s
[2/6] pnpm-0:10.9.0-1.fc42.noarch                             100% |   1.0 MiB/s |   3.1 MiB |  00m03s
[3/6] nodejs-full-i18n-1:22.22.0-2.fc42.x86_64                100% |   1.2 MiB/s |   8.5 MiB |  00m07s
[4/6] nodejs-npm-1:10.9.4-1.22.22.0.2.fc42.x86_64             100% |   1.3 MiB/s |   2.3 MiB |  00m02s
[5/6] nodejs-docs-1:22.22.0-2.fc42.noarch                     100% | 548.4 KiB/s |   9.1 MiB |  00m17s
[6/6] nodejs-libs-1:22.22.0-2.fc42.x86_64                     100% | 773.6 KiB/s |  19.6 MiB |  00m26s
--------------------------------------------------------------------------------------------------------
[6/6] Total                                                   100% | 439.5 KiB/s |  42.8 MiB |  01m40s
Running transaction
[1/8] Verify package files                                    100% |  20.0   B/s |   6.0   B |  00m00s
[2/8] Prepare transaction                                     100% |   5.0   B/s |   6.0   B |  00m01s
[3/8] Installing nodejs-libs-1:22.22.0-2.fc42.x86_64          100% |  67.8 MiB/s |  78.1 MiB |  00m01s
[4/8] Installing nodejs-1:22.22.0-2.fc42.x86_64               100% |   2.0 MiB/s | 163.7 KiB |  00m00s
[5/8] Installing pnpm-0:10.9.0-1.fc42.noarch                  100% |   6.6 MiB/s |  16.5 MiB |  00m03s
[6/8] Installing nodejs-full-i18n-1:22.22.0-2.fc42.x86_64     100% |  46.9 MiB/s |  30.4 MiB |  00m01s
```

Устаналиваю через pnpm commitizen и standard-changelog.

```
Update available! 10.9.0 → 10.30.3.
Changelog: https://pnpm.io/v/10.30.3
  To update, run: pnpm self-update

 WARN  2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +150
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
Progress: resolved 150, reused 0, downloaded 150, added 150, done

/home/gracimildevnanque/.local/share/pnpm/global/5:
+ commitizen 4.3.1

Done in 22s using pnpm v10.9.0
gracimildevnanque@fedora:~$ pnpm add -g strandarg-changelog
 ERR_PNPM_FETCH_404  GET https://registry.npmjs.org/strandarg-changelog: Not Found - 404

This error happened while installing a direct dependency of /home/gracimildevnanque/.local/share/pnpm/global/5

strandarg-changelog is not in the npm registry, or you have no permission to fetch it.

No authorization header was set for the request.
Progress: resolved 1, reused 1, downloaded 0, added 0
gracimildevnanque@fedora:~$ pnpm add -g strandard-changelog
 ERR_PNPM_FETCH_404  GET https://registry.npmjs.org/strandard-changelog: Not Found - 404

This error happened while installing a direct dependency of /home/gracimildevnanque/.local/share/pnpm/global/5

strandard-changelog is not in the npm registry, or you have no permission to fetch it.

No authorization header was set for the request.
Progress: resolved 1, reused 1, downloaded 0, added 0
gracimildevnanque@fedora:~$ pnpm add -g standard-changelog
 WARN  2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +30
++++++++++++++++++++++++++++++
Progress: resolved 180, reused 150, downloaded 30, added 30, done

/home/gracimildevnanque/.local/share/pnpm/global/5:
+ standard-changelog 7.0.1

Done in 7.3s using pnpm v10.9.0
gracimildevnanque@fedora:~$
```

Создаю новый репозиторий и делаю там первый коммит.

```
gracimildevnanque@fedora:~$ gh repo create --public git-extended
✓ Created repository Gracimilde/git-extended on github.com
  https://github.com/Gracimilde/git-extended
gracimildevnanque@fedora:~$ git clone --recursive git@github.com:Gracimilde/git-extended.git
Clonando para 'git-extended'...
warning: Pareces ter clonado um repositório vazio.
gracimildevnanque@fedora:~$ ls
'Área de Trabalho'   Documentos   gitflow   Modelos   Público          Vídeos
'Capturas de ecrã'   git-extended   Imagens   Música   Transferências   work
gracimildevnanque@fedora:~$ cd git-extended/
gracimildevnanque@fedora:~/git-extended$ touch README.md && echo text > README.md
gracimildevnanque@fedora:~/git-extended$ git add .
gracimildevnanque@fedora:~/git-extended$ git commit -m "frist commit"
[main (memória-raiz) 816879e] frist commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
gracimildevnanque@fedora:~/git-extended$ git push
Enumerating objects: 3, concluído.
A contar objetos: 100% (3/3), concluído.
Writing objects: 100% (3/3), 871 bytes | 871.00 KiB/s, concluído.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Gracimilde/git-extended.git
 * [new branch]      main -> main
gracimildevnanque@fedora:~/git-extended$
```

Рис. 3: Создание репозитория

Инициализирую и конфигурирую общепринятые коммиты в созданной директории через редактирование package.json.

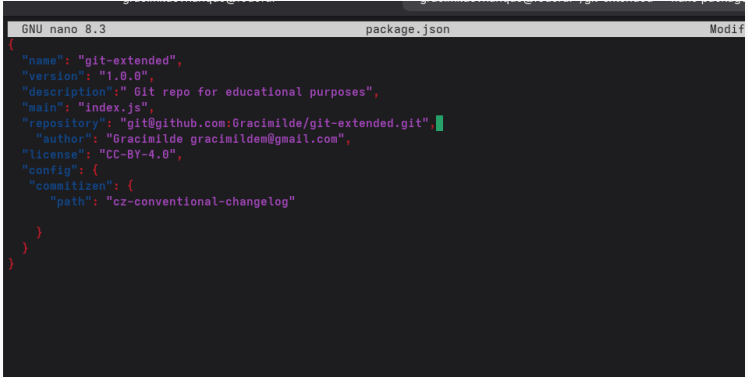

Рис. 4: Конфигурация package.json

Делаю снимок измененний, создаю коммит и отправляю на удаленный репозиторий.

```
gracimildevnanque@fedora:~/git-extended$ nano package.json
gracimildevnanque@fedora:~/git-extended$ git add .
gracimildevnanque@fedora:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: docs:     Documentation only changes
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 94 chars):
 (21) improve informatation
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[main 1e8cbe4] docs: improve informatation
 1 file changed, 16 insertions(+)
 create mode 100644 package.json
gracimildevnanque@fedora:~/git-extended$ git push
Enumerating objects: 4, concluído.
A contar objetos: 100% (4/4), concluído.
Delta compression using up to 7 threads
Comprimindo objetos: 100% (3/3), concluído.
Writing objects: 100% (3/3), 1.14 KiB | 1.14 MiB/s, concluído.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Gracimilde/git-extended.git
   816879e..1e8cbe4  main -> main
gracimildevnanque@fedora:~/git-extended$
```

Рис. 5: Отправление коммита на GitHub

Инициализирую в репозитории git flow и создаю 1 релиз в только что созданной ветке develop.

```
gracimildevnanque@fedora:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
   - main
Branch name for production releases: [main]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/gracimildevnanque/git-extended/.git/hooks]
gracimildevnanque@fedora:~/git-extended$ git branch
* develop
  main
gracimildevnanque@fedora:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/Gracimilde/git-extended/pull/new/develop
remote:
To github.com:Gracimilde/git-extended.git
 * [new branch]      develop -> develop
gracimildevnanque@fedora:~/git-extended$ git branch --set-upstream-to=origin/develop develop
ramo 'develop' configurado para monitorizar 'origin/develop'.
gracimildevnanque@fedora:~/git-extended$ git branch --set-upstream-to=origin/develop
ramo 'develop' configurado para monitorizar 'origin/develop'.
gracimildevnanque@fedora:~/git-extended$ git branch --set-upstream-to=origin/develop develop
ramo 'develop' configurado para monitorizar 'origin/develop'.
gracimildevnanque@fedora:~/git-extended$ git flow release start 1.0.0
Trocou para um novo ramo 'release/1.0.0'
```

Создаю список изменений через standard changelog, заканчиваю релиз и выгружаю на удаленный репозиторий изменения.

```
gracimildevnanque@fedora:~/git-extended$ git flow release finish 1.0.0
Branches 'main' and 'origin/main' have diverged.
And local branch 'main' is ahead of 'origin/main'.
Já em 'main'
Teu ramo está à frente de 'origin/main' por 2 memórias.
  (usa "git push" para publicar tuas memórias locais)
Trocou para o ramo 'develop'
Teu ramo está atualizado com 'origin/develop'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 CHANGELOG.md
Ramo release/1.0.0 eliminado (era 5fcd81a).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'main'
- The release was tagged 'v1.0.0'
- Release tag 'v1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

gracimildevnanque@fedora:~/git-extended$ git push --all
Enumerating objects: 6, concluído.
A contar objetos: 100% (6/6), concluído.
Delta compression using up to 7 threads
Comprimindo objetos: 100% (4/4), concluído.
Writing objects: 100% (5/5), 2.66 KiB | 1.33 MiB/s, concluído.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Gracimilde/git-extended.git
   1e8cbe4..9a8ae40  develop -> develop
```

Инициализирую ветку feature для работы над новой функциональностью, готовлю релиз и загружаю на github.

```
^LLLgracimildevnanque@fedora:~/git-extended$ git flow release start 1.2.3
Trocou para um novo ramo 'release/1.2.3'

Summary of actions:
- A new branch 'release/1.2.3' was created, based on 'develop'
- You are now on branch 'release/1.2.3'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

     git flow release finish '1.2.3'

gracimildevnanque@fedora:~/git-extended$ nano package.json
gracimildevnanque@fedora:~/git-extended$ standard-changelog
gracimildevnanque@fedora:~/git-extended$ git add .
gracimildevnanque@fedora:~/git-extended$ git status
Em ramo release/1.2.3
Alterações para serem memorizadas:
  (usa "git restore --staged <ficheiro>..." para desencenar)
        modificado:    CHANGELOG.md
        modificado:    package.json

gracimildevnanque@fedora:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip) site
? Write a short, imperative tense description of the change (max 87 chars):
 (16) update changelog
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[release/1.2.3 bba2685] chore(site): update changelog
 2 files changed, 3 insertions(+), 1 deletion(-)
gracimildevnanque@fedora:~/git-extended$ git flow release finish 1.2.3
Trocou para o ramo 'main'
Teu ramo está atualizado com 'origin/main'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 2 ++
 package.json | 2 +-
 2 files changed, 3 insertions(+), 1 deletion(-)
Já em 'main'
Teu ramo está à frente de 'origin/main' por 3 memórias.
```

Выводы

В ходе выполнения лабораторный работы я получил навыки правильной работы с репозиториями git.